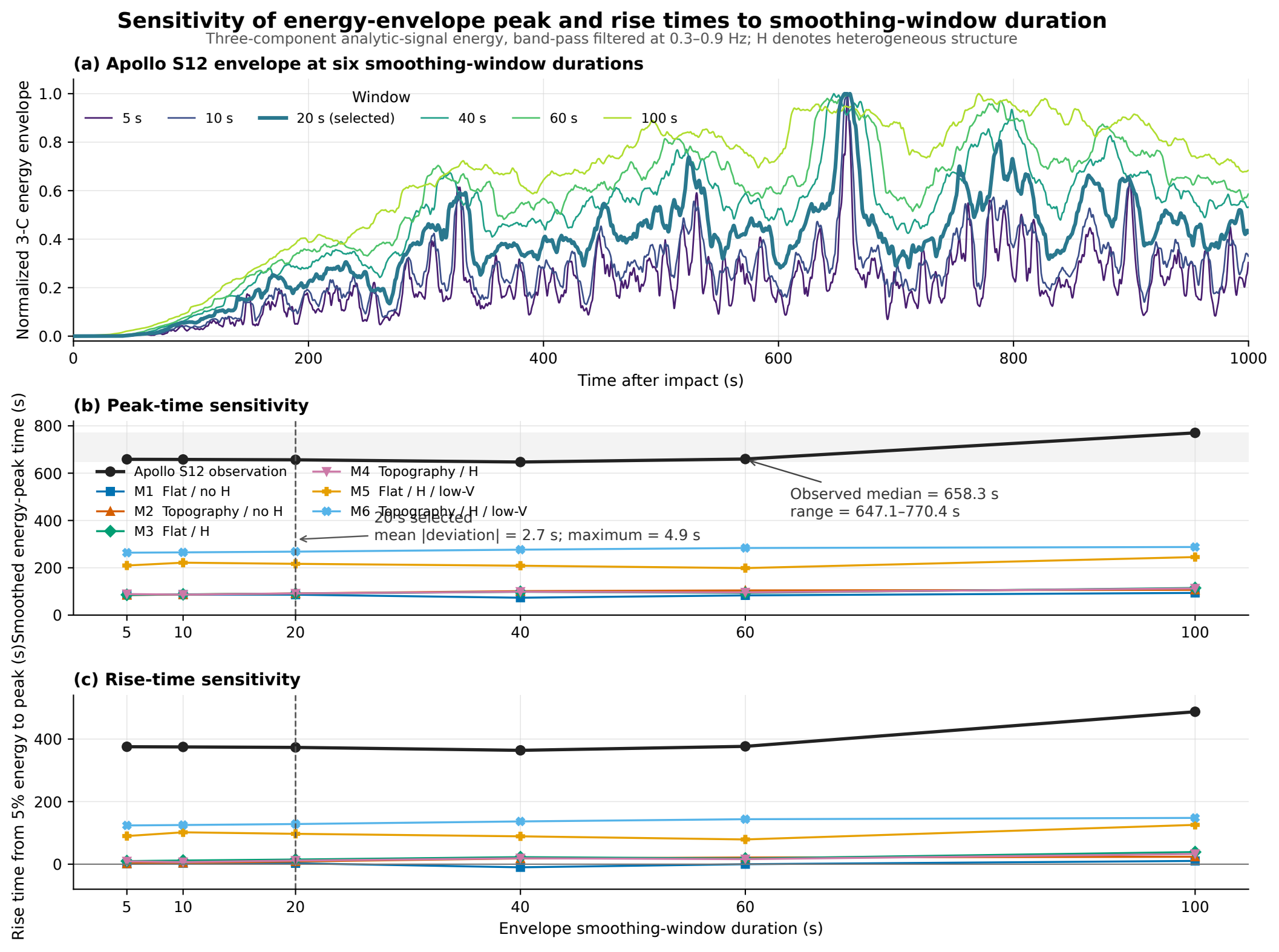

Sensitivity of energy-envelope peak and rise times to smoothing-window duration
Three-component analytic-signal energy, band-pass filtered at 0.3–0.9 Hz; H denotes heterogeneous structure
(a) Apollo S12 envelope at six smoothing-window durations
Window
5 s
10 s
20 s (selected)
40 s
60 s
100 s
Normalized 3-C energy envelope
Time after impact (s)
(b) Peak-time sensitivity
Apollo S12 observation
M1 Flat / no H
M2 Topography / no H
M3 Flat / H
M4 Topography / H
M5 Flat / H / low-V
M6 Topography / H / low-V
20 s selected
mean |deviation| = 2.7 s; maximum = 4.9 s
Observed median = 658.3 s
range = 647.1–770.4 s
Smoothed energy-peak time (s)
(c) Rise-time sensitivity
Rise time from 5% energy to peak (s)
Envelope smoothing-window duration (s)